梅思远

计算机科学博士生 | 医学人工智能与计算机视觉

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Pattern Recognition Lab (LME), FAU Erlangen–Nürnberg, Germany

研究方向

- 深度探索医学人工智能与计算机视觉领域的多个技术方向，研究图像生成、重建、分割与配准；方法上重点关注扩散/流匹配生成式模型、基础模型、表征学习与智能体工作流。除此之外，我一直保持对先进的通用语言模型和多模态智能体的持续关注和学习。

个人优势

- **科研闭环能力**：能够从问题定义、方法建模、代码实现和集群实验推进到论文写作与开源发布。第一作者成果覆盖多篇顶刊顶会oral和国际挑战赛第一，研究主题横跨基座模型的迁移学习、生成建模与多模态医学图像处理。
- **注重效率和目标导向工作思维**：擅长规划作业，设立并高效地完成阶段性目标，确保最终项目的顺利提交。
- **训练有素的组织与表达能力**：长期负责实验室组织任务，担任两门研究生主课的讲授和组织工作（学生达到300人以上）；具有多次国际会议的演讲经历。

教育经历

- **FAU Erlangen–Nürnberg** — 2023.09–2027.05（预计）
  计算机科学博士，ERC 项目全额资助；导师：Prof. Dr.-Ing. habil. Andreas Maier — 德国埃尔朗根
- **北京大学医学部** — 2026.05–2026.10
  访问学者；导师：Prof. Dr.-Ing. Yixing Huang — 中国北京
- **FAU Erlangen–Nürnberg** — 2020.09–2023.06
  医学工程硕士；毕业论文成绩1.0 — 德国埃尔朗根
- **浙江科技大学** — 2016.09–2020.06
  自动化专业工学学士；中德2+3 项目；优秀毕业生奖学金 — 中国杭州

专业经历

- **Friedrich-Alexander-Universität Erlangen–Nürnberg (FAU)** — *2023–2027*
  计算机科学博士生，Pattern Recognition Lab；导师：Prof. Andreas Maier — 德国埃尔朗根
  - 依托欧洲研究委员会（ERC）4D Nanoscope 项目，面向高分辨率X-ray 显微成像与光片荧光显微成像，设计并实现高效配准流程，重点解决大体积、多模态数据下的配准效率与稳定性问题；成果发表于JMI 并获**Featured Paper**。
  - 对图像生成方向进行深入研究。最近MICCAI成果围绕扩散/流模型在医学图像转换中的多步采样与随机性问题，研究一步生成建模，在保持单步推理效率的同时提升图像转换质量；成果获**MICCAI 2026 Early Accept + Oral Presentation**。
  - 对视觉基座模型的下游迁移能力进行扩展。研究如何将预训练Vision Transformer 的中间表征有效用于分割等稠密预测任务，形成通用的轻量调节器机制；第一作者论文发表于**Pattern Recognition**。
  - 参与多个国际性挑战赛。开发多模态到CT生成模型，作为主要完成人获得**MICCAI SynthRAD2025 Task 2 第一名、Task 1 第四名**。
  - 掌握各种成像以及重建技术。相关论文发表ISBI 2023, CT Meeting 2024会议与**TMI**, PMB, BMPC等期刊。
  - 讲授和组织多门研究生主课，包括*Flat-panel CT Reconstruction* 和*Vibe Coding: Software Engineering with LLMs* 新课开发。
- **北京大学医学部Artificial Intelligence for Medical Imaging (AIMI) Lab** — *2026.05–2026.10*
  访问学者；导师：Prof. Yixing Huang — 中国北京
  - 参加MICCAI BIC-MAC | PET 衰减校正挑战赛，面向多模态输入到合成CT 的生成任务，服务于PET attenuation correction；在验证阶段CT 指标取得第一名，并围绕sCT、μ-map 与PET 重建指标之间的误差传递开展系统诊断。
  - 和实验室交流，深化对CT/CBCT 成像链路及临床应用的理解；期间访问重庆ICT Center，并参加清华大学MISC 2026。
- **NLP与Agent开源项目** — *2026.8–present*
  - 自我钻研并精细理解主流LLM的技术报告以及Stanford CS336, CS224N, CS234系列课程。掌握GPT, QWEN, Deepseek等主流模型开源技术细节。完成nano-GPT的训练项目。长期使用agent（Codex，Claude Code）工作。

精选论文

J=期刊，C=会议，M=手稿

[C.1] **Siyuan Mei**, Yanteng Zhang, Yan Xia, Qizhen Lan, Yipeng Sun, Siming Bayer, Zirong Li, Chengze Ye, Daiqi Liu, Xiaoqian Jiang, Fuxin Fan, Yixing Huang, and Andreas Maier. *Time Matters: Rethinking Diffusion and Flow Models in One-Step Medical Image Translation*. **MICCAI**, **2026** (**Early Accept; Oral Presentation**).

[C.2] **Siyuan Mei**, Yan Xia, Yipeng Sun, Siming Bayer, Zirong Li, Chengze Ye, Daiqi Liu, Fuxin Fan, Yixing Huang, and Andreas Maier. *WING: A Window-Prior-Based Generative Network with Gated Inception for Cross-Modality CT Synthesis*. **MICCAI MIART Workshop**, **2026**.

[C.3] Daiqi Liu, Lukas Mulzer, Md Hasan, Nyvenn de Castro, Fangxu Xing, Xingjian Kang, Chengze Ye, **Siyuan Mei**, Yipeng Sun, Tomás Arias-Vergara, Jana Hutter, Jonghye Woo, Andreas Maier, and Paula Andrea Pérez-Toro. *Speech-Guided Multimodal Learning for Vocal Tract Segmentation in Real-Time MRI*. **MICCAI**, **2026** (**Early Accept; Oral Presentation**).

[C.4] Mengchen Fan, Baocheng Geng, Xi Xiao, Tianyang Wang, **Siyuan Mei**, Pulin Che, Xiaoqian Jiang, and Qizhen Lan. *Detail Consistent Stage-Wise Distillation for Efficient 3D MRI Segmentation*. **MICCAI**, **2026** (**Early Accept; Spotlight Presentation**).

[J.1] **Siyuan Mei**, Mareike Thies, Yan Xia, Yipeng Sun, Fei Wu, Fuxin Fan, Mingxuan Gu, Chengze Ye, Yixing Huang, Vincent Christlein, and Andreas Maier. *Vision Transformer Hook for Dense Predictions*. **Pattern Recognition**, **179 (2026), 113818**.

[J.2] **Siyuan Mei**, Fuxin Fan, Mareike Thies, Mingxuan Gu, Fabian Wagner, Oliver Aust, Ina Erceg, Zeynab Mirzaei, Georgiana Neag, Yipeng Sun, Yixing Huang, and Andreas Maier. *BigReg: An Efficient Registration Pipeline for High-Resolution X-Ray and Light-Sheet Fluorescence Microscopy*. **Journal of Medical Imaging**, **12(5), 054004 (2025)** (**Featured Paper**).

[C.5] **Siyuan Mei**, Fuxin Fan, and Andreas Maier. *Metal Inpainting in CBCT Projections Using Score-based Generative Model*. **IEEE ISBI**, **2023**.

[J.3] Mareike Thies, Fabian Wagner, Noah Maul, Haijun Yu, Manuela Goldmann, Linda-Sophie Schneider, Mingxuan Gu, **Siyuan Mei**, Lukas Folle, Alexander Preuhs, Michael Manhart, and Andreas Maier. *A Gradient-Based Approach to Fast and Accurate Head Motion Compensation in Cone-Beam CT*. **IEEE Transactions on Medical Imaging**, **44(2), 1098–1109 (2025)**.

[J.4] Chengze Ye, Linda-Sophie Schneider, Yipeng Sun, Mareike Thies, **Siyuan Mei**, Paula Andrea Pérez-Toro, Siming Bayer, and Andreas Maier. *Robustness and Stability Analysis of Differentiable Shift-Variant FBP for Cone-Beam CT under Challenging Acquisition Settings*. **Machine Learning for Biomedical Imaging**, **2026, 284–296**.

[J.5] Yipeng Sun, Linda-Sophie Schneider, **Siyuan Mei**, Jinhua Wang, Ge Hu, Mingxuan Gu, Chengze Ye, Fabian Wagner, Lan Song, Siming Bayer, and Andreas Maier. *Filter2Noise: A Framework for Interpretable and Zero-Shot Low-Dose CT Image Denoising*. **Journal of Medical Imaging**, **13(2), 024004 (2026)**.

[J.6] Chengze Ye, Linda-Sophie Schneider, Yipeng Sun, Mareike Thies, **Siyuan Mei**, and Andreas Maier. *DRACO: Differentiable Reconstruction for Arbitrary CBCT Orbits*. **Physics in Medicine and Biology**, **70, 075005 (2025)**.

[C.6] Mareike Thies, Noah Maul, **Siyuan Mei**, Laura Pfaff, Nastassia Vysotskaya, Mingxuan Gu, Jonas Utz, Dennis Possart, Lukas Folle, Fabian Wagner, and Andreas Maier. *Differentiable Score-Based Likelihoods: Learning CT Motion Compensation From Clean Images*. **MICCAI**, **2024**.

[C.7] Mingxuan Gu, Mareike Thies, **Siyuan Mei**, Fabian Wagner, Mingcheng Fan, Yipeng Sun, Zhaoya Pan, Sulaiman Vesal, Ronak Kosti, Dennis Possart, Jonas Utz, and Andreas Maier. *Unsupervised Domain Adaptation Using Soft-Labeled Contrastive Learning with Reversed Monte Carlo Method for Cardiac Image Segmentation*. **MICCAI**, **2024**.

[C.8] Laura Pfaff, Fabian Wagner, Nastassia Vysotskaya, Mareike Thies, Noah Maul, **Siyuan Mei**, Tobias Wuerfl, and Andreas Maier. *No-New-Denoiser: A Critical Analysis of Diffusion Models for Medical Image Denoising*. **MICCAI**, **2024**.

[M.1] Zirong Li, **Siyuan Mei**, Weiwen Wu, Andreas Maier, Lina Gölz, and Yan Xia. *Generative Drifting for Conditional Medical Image Generation*. **arXiv preprint**, **2026**.

[M.2] **Siyuan Mei**, Yan Xia, and Fuxin Fan. *GANeXt: A Fully ConvNeXt-Enhanced Generative Adversarial Network for MRI- and CBCT-to-CT Synthesis*. **arXiv preprint (Challenge Report)**, **2025**.

荣誉与奖励

- **MICCAI SynthRAD2025 Challenge：Task 2 第一名、Task 1 第四名** *2025*
- **FAU Excellent International Graduate Student Scholarship** *2023*
- **FAU Excellent MRI Course Achievement Scholarship** *2022*
- 浙江科技大学优秀毕业生奖学金 *2020*

教学经历

- **Flat-panel CT Reconstruction** *2024–2026 夏季学期*
  FAU Erlangen–Nürnberg
- **Vibe Coding: Software Engineering with LLMs** *2026 夏季学期*
  FAU Erlangen–Nürnberg；参与课程讲义与Springer 图书配套材料建设。

学术服务

- 期刊审稿：Pattern Recognition；Journal of Machine Learning for Biomedical Imaging (MELBA)。
- 技术编辑：Journal of Medical Imaging。
- 会议审稿：MICCAI。

其他信息

- 语言：中文（母语）、英语（专业工作交流）、德语（C1 证书）。

- **兴趣**：健身、网球（Level 2.5）。
- **MBTI**：ENTJ-A。
- **求职意向**：医学影像、计算机视觉、多模态生成模型与智能体开发。